

TASIN ISLAM

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SUMMARY

- Highly knowledgeable and skilled AI engineer seeking full-time opportunities in AI engineering or a related field.
- Proficient in developing popular computer vision models such as GANs, diffusion models and ViTs using popular libraries like PyTorch and Tensorflow.
- Possess advanced capabilities in using open-source LLM orchestration libraries like LangChain to develop RAG and other NLP applications.
- Proven ability to manage time effectively and deliver under pressure, evident from handling project work alongside a teaching role at Brunel University.
- Possess strong programming skills in Python, Java and C++ acquired through professional work experience and university assignments.

EDUCATION

Oct 2020 - Oct 2024	PhD Computer Science	Brunel University
Sept 2017 - July 2020	BSc Computer Science (AI) First Class Honours	Brunel University

SKILLS

Programming: Python, Java, C++, R
Database: SQL, MySQL, Databricks SQL,
AI/ML/Data Science: PyTorch, Scikit-learn, PySpark, Tensorflow, MLflow, NumPy, Pandas, OpenCV, Hugging
Spacy, Sentence Transformers, FAISS, LLMs, LangChain, RAG, Copilot Studio, Agentic AI,
Prompt Engineering
Tools/IDEs: PyCharm, Jupyter Notebook, Visual Studio, Eclipse, RStudio, Docker, Databricks,
Azure Data Lake, Azure Data Factory
OS: Windows, Linux, Mac
Mobile development: Android
Data Representation: XML, JSON
Data visualisation: MS Excel, Plotly, Matplotlib, Plotly Dash, Tableau, PowerBI
Version Control: Git, GitHub, Azure DevOps, GitKraken

EMPLOYMENT HISTORY

Feb 2024 – Present	Data Scientist	Metropolitan Police
• Developed and implemented advanced algorithms in computer vision, NLP, and other data-driven techniques to derive meaningful value from data to support police operations. • Developed advanced GenAI techniques, including RAG with vector databases and GraphRAG, to retrieve relevant information from big data that support officers in policing operations. • Utilised Copilot Studio to develop Agentic AI, which addresses various issues that an officer may encounter. It can perform web scraping, document lookups, and link various sources. • Leveraged statistical techniques and machine learning to measure the professionalism and performance of police officers. • Utilised industry-standard tools like Azure and Databricks to deploy AI/ML products for client use. • Productionised projects by developing web applications using Plotly Dash and used Databricks services like MLflow for product deployment and monitoring. • Knowledgeable about deployment environments and the process of transitioning projects from development to production, including the use of containerisation tools like Docker and environmental labels. • Contributed to the development of more effective and proactive policing strategies through data-driven insights.		

Sept 2020 – Nov 2023	Doctoral Researcher	Brunel University
• Conducting original and in-depth research specifically focused on virtual fitting rooms (VFR), which aim to facilitate virtual interactions between consumers and products offered by online fashion businesses. • Explored the latest advancements in computer vision and deep learning techniques to develop innovative approaches for VFR applications. • Contributing new knowledge and insight by proposing novel deep learning models that outperform previous methods. • Published research findings in reputable conferences and journals, showcasing the contributions made to the academic community and the wider industry. • Presented research work at conferences and symposiums, fostering networking opportunities and receiving valuable feedback from peers and experts in the field.		
Sept 2020 – Present	Graduate Teaching Assistant	Brunel University
• Assisted and supported students in completing lab work, providing guidance and answering questions related to the course material. • Offered technical support, including software installation and code debugging, to ensure students' smooth progress in their assignments. • Explained the functionality and purpose of specific lines of code, helping students understand the underlying concepts. • Supported modules such as data analysis using R and RStudio, networking (covering threads and concurrent control in Java), and conducted assessments to evaluate students' code functionality and comprehension.		
Jan 2022 – Sept 2022	System Administrator	Brunel University
• Installed vital software packages and drivers on two Ubuntu GPU clusters, ensuring seamless functionality. • Managed and controlled job allocation using the Slurm job scheduler, optimising resource utilisation and facilitating fair access for users. • Collaborated with on-site teams to troubleshoot technical issues, minimising downtime and maintaining high system availability. • Conducted regular system monitoring, performance analysis, and capacity planning to identify bottlenecks and implement proactive solutions. • Assisted users with cluster utilisation, providing technical support and guidance on job submission, troubleshooting, and optimisation.		
Dec 2021 – Sept 2022	Senior Research Assistant	Brunel University
• Executed experiments on hybrid agent-based and discrete-event simulation software to predict the number of COVID-19 infections in a specified area, utilising various parameters such as setting lockdowns, infection rates, and population demographics. • Conducted multiple runs of the experiments with different parameter configurations to analyse and understand the impact of various factors on the spread of COVID-19. • Utilised data visualisation tools such as Matplotlib to plot graphs comparing the number of infections across different regions and other parameters, providing clear visual representations of the simulation results. • Presented experiment results to colleagues, effectively communicating the findings and explaining the insights obtained from the graphs and data analysis. • Actively engaged with the developers of the simulation software, providing feedback and notifying them of any unreasonable results or issues encountered during the experiments.		

PROJECTS

Triaging FOI request with LLM

Metropolitan Police

- Processed and analysed a high volume of daily FOI requests received by the police organisation, identifying a significant number of non-legitimate submissions.
- Utilised historical records containing labelled datasets of valid and invalid FOI requests to develop a classification model.
- Fine-tuned a GPT-2 model on the historical dataset to automate the classification of FOI requests as valid or invalid.
- Improved efficiency for public relations and data rights teams by enabling them to focus solely on legitimate FOI requests, optimising workflow and resource allocation.

Firearm Detection (github.com/tasinislam21/gun_detection_python_vs_c_plus)

Metropolitan Police

- Utilised YOLOv3 to effectively identify and draw bounding boxes around firearms in live-stream video footage, enhancing real-time threat detection capabilities.
- Conducted extensive experimentation with both Python and C++ to compare computational efficiency and resource demands.
- Successfully productionised the model using Docker to ensure smooth deployment on Azure.
- Leveraged Gradio to create user-friendly interfaces, deploying the model as Databricks apps.

Supporting intelligence gathering with GraphRAG

Metropolitan Police

- Developed a GenAI chatbot to streamline intelligence gathering for officers overwhelmed by numerous internal policies.
- Designed a solution that interprets officers' initial descriptions and retrieves relevant policy guidance.
- Utilised a GraphRAG approach: transformed internal policy documents into a Neo4j-based knowledge graph.
- Employed LangChain's GraphRAG library to generate Cypher queries using the LLaMA model, enabling precise extraction of relevant policy nodes.
- Chatbot provides real-time instructions and recommendations aligned with internal protocols, improving operational efficiency and compliance.

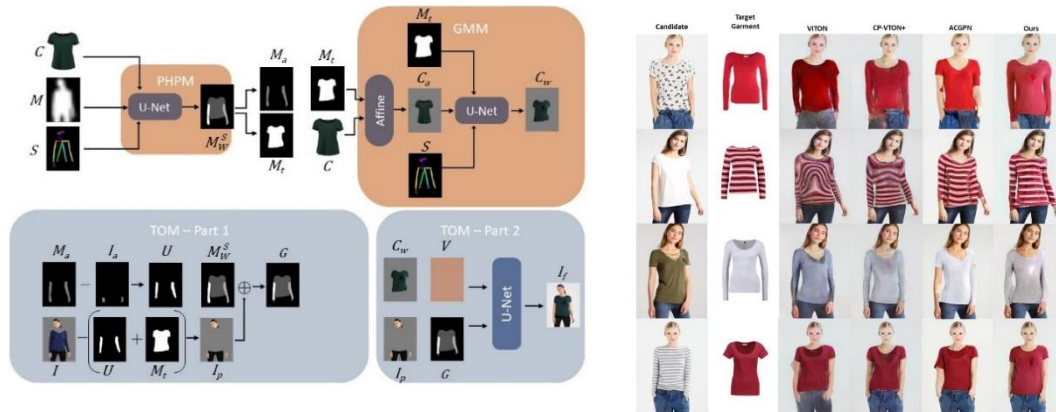
Fleet Simulation

Metropolitan Police

- Developed a discrete event simulation (DES) model to simulate the movement of fleet vehicles.
- Enabled the fleet management team to experiment with resources such as the number of vehicles or driver officers.
- Demonstrated how long it takes a unit to arrive at the scene.
- Illustrated potential impacts of insufficient vehicles or police drivers on arrival times.
- Simulated the number of incidents occurring in a 24-hour period, reflecting low incidents in the morning and high incidents during the day/evening.

Image-based Virtual Try-On (github.com/1702609/SVTON)

- Developed a novel approach utilising Generative Adversarial Network (GAN) to replace a person's garment with a desired item.
- Contributions include refining the candidate representation to incorporate enhanced posture information, implementing a truncated U-Net for improved performance, and utilising Affine Transform for efficient garment transformation.
- Evaluated the performance of the developed model by comparing it to state-of-the-art models in the field. The results showcased superior qualitative and quantitative outcomes, highlighting the model's ability to generate realistic and high-quality try-on images.



Multi-Pose Virtual Try-On (github.com/1702609/multi_pose_vton)

- Incorporated a unique feature into the traditional virtual try-on model, enabling the ability to change the posture of the person while simultaneously replacing the garment.
- Contributed to the development of a pipeline that integrates the pose transfer module, based on a StyleGAN architecture, with a traditional virtual try-on framework primarily utilising U-Net.
- Demonstrated the superiority of the model by synthesising images of significantly higher quality compared to previous works. The generated images reached a supreme level of realism, surpassing the limitations of prior research efforts.

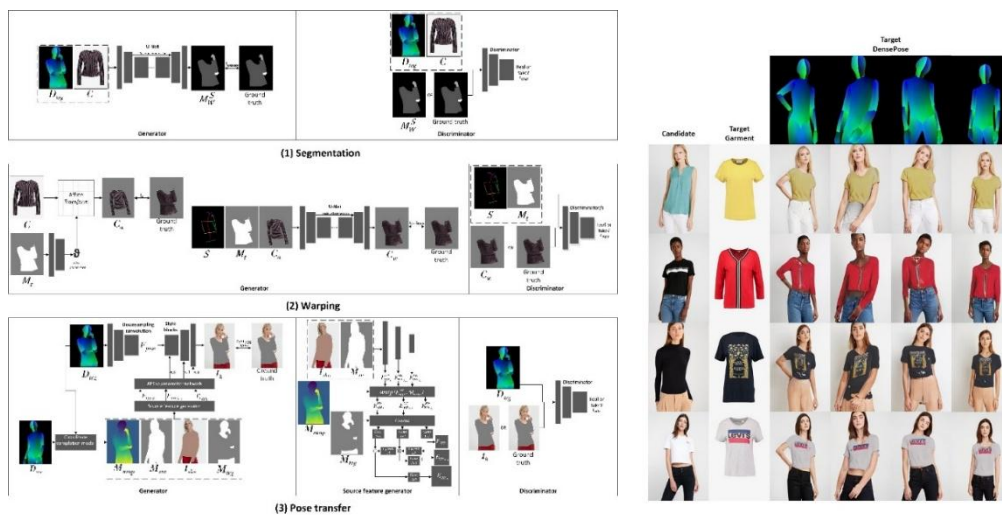
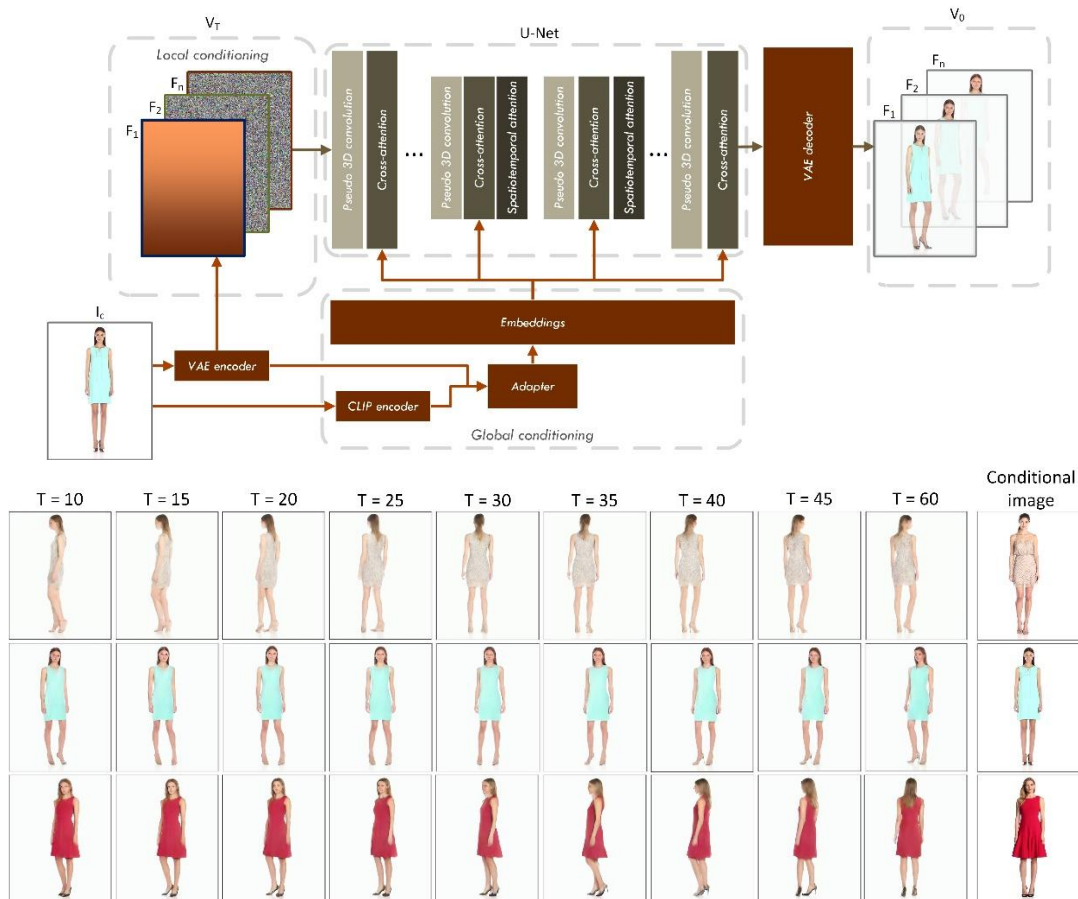


Image-to-Video Synthesis (github.com/1702609/FashionFlow) or (huggingface.co/spaces/sunjuice/FashionFlow)

- Developed a generative model for synthesising videos performing random yet plausible movements from fashion images.
- Leverages latent video diffusion techniques, comprising pseudo-3D convolutional layers designed to optimise efficiency and ease of training, a cross-attention mechanism that effectively incorporates the fashion image into the video generation process whilst being accompanied by an adapter module to improve the efficacy.
- This model significantly outperformed our competitors by producing smoother and better-conditioned videos. It enhances the potential shopping experience.



Final Year Project (github.com/1702609/FYP/tree/master/multiPerson)

- Developed a Python solution that utilises pose estimation to recognise instances of physical crimes in CCTV footage.
- Created a rule-based AI system capable of identifying physical contact between individuals and recalling the location of the attack.
- Analysed complex problems and effectively broke them down into smaller sections, facilitating the efficient development of the software.

Multi-player Yahtzee game (github.com/1702609/yahtzee)

- Developed a multiplayer Yahtzee game in Java that enables 3 or more clients to play together.
- Implemented multithreading on the server side to manage and synchronise multiple connected clients.
- Designed and developed the server-client communication protocol, ensuring reliable data transmission and handling potential network issues.

- Implemented game mechanics such as rolling dice, scoring, and tracking player turns, ensuring accurate gameplay according to the Yahtzee rules.
- Utilised concurrent control to handle scoreboard requests and updates from clients, preventing multiple client threads from modifying the scoreboard simultaneously.

Group Project – Android Development

- Contributed to a university team project with six members, collaborating on the development of an Android dating app.
- Actively participated in brainstorming sessions and feature planning meetings, offering creative ideas and suggestions to enhance the app's functionality and user experience.
- Worked closely with team members to assign tasks, set milestones, and establish a collaborative development approach, ensuring efficient progress and timely completion of the project.
- Effectively communicated with team members to establish a collaborative development approach and fostered trust, resulting in an enjoyable project completion experience.
- Took ownership of specific modules within the app, developing clean and well-structured code while adhering to best practices and coding standards.
- Conducted thorough testing when merging code with other team members, working together to ensure the app was functioning properly and free of conflicts.
- Self-taught Android Programming through online resources, dedicating extensive hours to research and enhancing problem-solving skills for creating high-quality Android apps.

Interests

- Karate: Passionately engaged in Karate since 2010, earning a first-degree black belt. Actively involved in teaching new students, fostering focus and teamwork skills through participation in competitions.
- Gym: Regularly attend the gym to enhance concentration, stamina, and endurance, providing valuable support in handling stressful conditions.
- Reading: Devote time to reading a wide range of literature, expanding knowledge and fostering interest in storytelling and personal growth.

Awards and Publications

- Islam, T (2021, May). Deep Learning for Personalised Marketing. Computer Science Brunel PhD Symposium, 2021. **Awarded for best early PhD presentation.**
- Islam, T., Miron, A., Liu, X., & Li, Y. (2022, May). Svton: Simplified virtual try-on. Computer Science Brunel PhD Symposium, 2022. **Awarded for best PhD presentation.**
- Islam, T., Miron, A., Liu, X., & Li, Y. (2024). Image-based virtual try-on: Fidelity and simplification. Signal Processing: Image Communication, 129, 117189. (IF: 4.223)
- Islam, T., Miron, A., Nandy, M., Choudrie, J., Liu, X., & Li, Y. (2024). Transforming Digital Marketing with Generative AI. Computers, 13(7), 168. (IF: 3.938)
- Islam, T., Miron, A., Liu, X., & Li, Y. (2024). Dynamic Fashion Video Synthesis from Static Imagery. Future Internet, 16(8), 287. (IF: 4.318)
- Islam, T., Miron, A., Liu, X., & Li, Y. (2024). StyleVTON: A multi-pose virtual try-on with identity and clothing detail preservation. Neurocomputing, 594, 127887. (IF: 6.789)
- Islam, T., Miron A., Liu X., & Li, Y. (2024). Deep Learning in Virtual Try-On: A Comprehensive Survey. IEEE Access. (IF: 4.644)
- Islam, T., Miron A., Liu X., & Li, Y. (2022). SVTON: Simplified Virtual Try-On. In International Conference on Machine learning and Applications. (Acceptance Rate: 20%)
- Mintram, K., Anagnostou, A., Anokye, N., Okine, E., Groen, D., Saha, A., ... & Taylor, S. J. (2022). CALMS: Modelling the long-term health and economic impact of Covid-19 using agent-based simulation. Plos one, 17(8), e0272664. (IF: 3.107)
- Anagnostou, A., Groen, D., Taylor, S. J., Suleimenova, D., Abubakar, N., Saha, A., ... & Anokye, N. (2022). FACS-CHARM: a hybrid agent-based and discrete-event simulation approach for Covid-19 management at regional level. In 2022 Winter Simulation Conference (WSC) (pp. 1223-1234). IEEE.